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23 JULY 1962

SUPERSEDING
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MILITARY SPECIFICATION

CELLULOSE, WOODPULP (SULFATE)

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one grade of cellulose woodpulp sulfate for use in the manufacture of nitrocellulose.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-P-31 —Paper, General Specification and Method of Testing.
PPP-T-76 —Tape, Pressure-Sensitive Adhesive, Paper, Water-Resistant (for Carton Sealing).
UU-T-116 —Tape, Paper, Gummed, Water-Resistant.
UU-P-268 —Paper, Kraft, Wrapping.

MILITARY

MIL-A-140 —Adhesive, Water-Resistant, Waterproof Barrier Material.
MIL-N-244 —Nitrocellulose, for Use in Explosives.

STANDARDS

MILITARY

MIL-STD-105 —Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 —Inspection Terms and Definitions.
MIL-STD-129 —Marking for Shipment and Storage.

PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 —Procedures and Tables for Continuous Sampling by Attributes.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

TAPPI METHODS

Technical Association of the Pulp and Paper Industry.

(Application for copies should be addressed to the Secretary, TAPPI, 360 Lexington Ave., New York 17, N. Y.)

CONSOLIDATED CLASSIFICATION COMMITTEE

Consolidated Freight Classification.

(Application for copies should be addressed to the CONSOLIDATED CLASSIFICATION COMMITTEE, 202 Chicago Union Station, Chicago 6, Ill.)

FSC 6810

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AMERICAN TRUCKING ASSOCIATION PUBLICATION

National Motor Freight Classification Rules and Container Specifications.

(Application for copies should be addressed to the AMERICAN TRUCKING ASSOCIATION, 1424 16th St. N.W., Washington, D.C.)

3. REQUIREMENTS

3.1 Physical and chemical properties.

3.3.1 *Physical properties.* The woodpulp cellulose shall conform to the limits of the properties specified in table I when determined in accordance with the applicable subparagraphs specified herein.

TABLE I. *Physical properties*

Property	*Minimum (min.)*	*Maximum (max.)*	*Subparagraphs*
Inside diameter of core, inches (in.)	2.75	3.5	4.3.2.1
Length of sheets, feet (ft.)	1000	....	4.3.2.2
Width of sheets, in.	29.50	30.00	4.3.2.2
Outside diameter of rolls, in.	....	42.0	4.3.2.3
Moisture, percent	....	10.0	4.4.1
Density, ounce (oz.) (per sq. ft. per 0.01 in. thickness).	....	0.46	4.4.2
Thickness, in.	....	0.056	4.4.3
Weight (oz. per sq. ft.)	2.15	....	4.4.4
Bursting Strength (points)	....	125	4.4.5
Viscosity, centipoises	....	140	4.4.6

3.1.2 *Chemical properties.* The woodpulp cellulose shall conform to the limits of the properties specified in table II when determined in accordance with the applicable subparagraphs specified herein.

TABLE II. *Chemical properties*

Property	*Percent*		*Subparagraph*
	Minimum	*Maximum*	
Alpha cellulose	91.0	....	4.4.7
Ether extractive matter	...	0.30	4.4.8
Alkali soluble matter	...	10.00	4.4.9
Ash	...	0.30	4.4.10
Lignin	...	0.4	4.4.11
Silica	...	0.008	4.4.12
Pentosan	...	6.0	4.4.13

3.2 Workmanship.

3.2.1 *Rolls before wrapping.* The rolls of woodpulp cellulose shall be free from dirt, grit, water, oil, grease, and other foreign material.

3.3.2 *Rolls after wrapping.* The rolls of woodpulp cellulose and wrapping shall be free from water, oil, grease and other foreign material that may be absorbed by the wrapping.

3.3 **Nitration.** The woodpulp cellulose shall be nitrated to grade A type II or grade B nitrocellulose of Specification MIL–N–244 (see 4.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contractor or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define the terms used herein. Inspection shall be performed in accordance with this specification and other specifications referenced in any of the contractual documents.

4.1.1 *Contractor quality assurance system.* If the contractor desires to utilize a quality assurance system, which is at variance with the quality assurance provisions of 4.3 and 4.4 and other documents referenced herein, he shall submit a written description of the system to the contracting officer for approval prior to initiation of production. It shall include a description covering controls for lot formation and identification, inspections to be performed, inspection stations, sampling procedures, methods of inspection, (measuring and testing equipment), and provisions for control and disposition of nonconforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance provisions

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required by the provisions of 4.3 and 4.4 and the other documents referenced herein. The contractor shall not be restricted to the inspection station or the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedure of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any changes to the written procedure that effects the degree of assurance required by this specification or other documents referenced herein.

4.1.2 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product submitted:

(a) A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

(b) Quantity of product inspected.

(c) Results obtained for all inspection performed.

(d) Specification number and date, together with an identification and date of changes.

(e) Certificates of analysis on all material procured directly by the contractor when such material is covered by referenced Government specification.

(f) Quantity of product in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 *Government verification.* Using the contractor's written quality assurance procedure (see 4.1.1), this detail specification, and other contractual documents as a guide, the Government inspector shall verify all quality assurance operations performed by the contractor. Verification shall be in accordance with a or b as applicable, the decision being the responsibility of the procuring activity. In either case, the inspector shall also ascertain, prior to acceptance, that all quality assurance provisions of other specifications referenced in any of the contractual documents have been complied with. Deviations from prescribed or agreed upon procedures discovered by the Government inspector shall be brought to the attention of the supplier. Disposition of the product and remedial action shall be as directed by the Government inspector and, depending on the nature of the deviation, may consist of lot rejection, screening, resampling, reinstruction of the supplier's employees, or other appropriate action:

(a) Verification at the point of manufacture shall be accomplished at unscheduled intervals in accordance with 4.1.3.1 and 4.1.3.2.

(b) Verification at the point of delivery shall be in accordance with 4.1.3.2.

4.1.3.1 *Surveillance.* Surveillance shall include, but is not limited to:

(a) Observation of procedures concerning lot formation and identification.

(b) Observation of sampling procedures and application of acceptance criteria.

(c) Determination that all required examinations and tests are performed in accordance with the prescribed procedures of this specification, or approved equivalents thereto.

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(d) Review of procedures for control and disposition of nonconforming material.

4.1.3.2 *Product inspection.* Product inspection shall consist of Government inspection of product which has been previously inspected by the contractor and found to meet the quality assurance provisions of this specification. The inspection by the Government shall be performed in order to determine that the product is of the quality required by this specification and that the contractor's records are reliable.

4.2 Initial production.

4.2.1. *Submission.* The contractor shall submit an initial production sample consisting of 7 to 10 tons of woodpulp cellulose sulfate, which has been produced by the production process which the contractor will use in fulfilling the contract, to a Government approved facility designated by the contracting officer. An initial production sample shall also be submitted whenever a change occurs in the manufacturing process, material used, specification, or source of supply, such as to significantly affect product uniformity as determined by the Government.

4.2.2 *Inspections to be performed code No. 18.* The material shall be inspected for all the inspections specified in 4.3 and 4.4 of this specification. In addition, the following special examinations and tests shall be performed:

The woodpulp cellulose sulfate shall be shredded and mitrated to grade A, type II or grade B nitrocellulose of Specification MIL–N–244 to determine suitability in plant scale equipment.

4.2.3 *Rejection.* If the material fails to comply with the applicable requirements, the initial production quantity shall be rejected and the contractor shall take the necessary corrective action and submit a new initial production quantity, or portion thereof, as specified by the contracting officer. The contractor shall continue to submit new or supplemental quantities as necessary until such time as a quantity successfully passes the inspection specified or until otherwise directed by the procuring activity.

4.3 Inspection provisions.

4.3.1 *Lot formation.* A lot shall consist of continuous sheet rolls of sulfate woodpulp cellulose made from one or more batches of bleached sulfate woodpulp cellulose that have been subjected to the same processing operations and conditions, from one manufacturer, and in accordance with the same specification and same specification revision. A batch shall be that quantity of raw materials which has been subjected to the same unit chemical process or physical mixing process intended to make the final product homogenous.

4.3.2 *Examination.* Sampling plans and procedures for the following classification of defects shall be in accordance with Standard MIL–STD–105. Continuous sampling plans, in accordance with Handbook ORD–M608–11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect.

4.3.2.1 *Core (see 5.1 and 5.2).*

Categories	*Defects*	*Code No. 01 (see 6.3) Method of inspection*
Critical: None defined.		
Major: None defined.		
Minor—AQL 0.40 percent		
201.	Inside diameter ..... Gage	01001

4.3.2.2 *Sheets, prior to winding.*

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Categories	Defects	Code No. 02 Method of inspection	
Critical: None defined.			
Major—AQL 0.25 percent			
101.	Width	Gage	02001
Minor—AQL 1.00 percent			
201.	Length, min.	Gage	02002
202.	Evidence of poor workmanship (see 3.2)	Visual	02003

4.3.2.3 *Rolls, prior to wrapping (See 5.1 and 5.2).*

Categories	Defects	Code No. 03 Method of inspection	
Critical: None defined.			
Major: None defined.			
Minor—AQL 1.50 percent			
201.	Outside diameter of roll, max.	Gage	03001
202.	Core missing	Visual	03002
203.	Evidence of poor workmanship (see 3.2.1)	Visual	03003

4.3.2.4 *Rolls after wrapping and sealing (see 5.1 and 5.2).*

Categories	Defects	Code No. 04 Method of inspection	
Critical: None defined.			
Major: None defined.			
Minor—AQL 2.50 percent			
201.	Protective wrapping, torn, cut, perforated or incomplete	Visual	04001
202.	Adhesive missing, incomplete or improperly positioned	Visual	04002
203.	Tape incorrect type, missing, incomplete or improperly positioned	Visual	04003
204.	Markings incorrect, misleading or unidentifiable	Visual	04004
205.	Evidence of poor workmanship (see 3.2.2)	Visual	04005

4.3.3 *Testing.*

4.3.3.1 *Sampling.*

4.3.3.1.1 *From the lot.* Five rolls shall be selected, at random, from the lot.

4.3.3.1.1.1 *For determination of chemical properties.* Samples shall be taken by boring into each roll to a depth of 3 inches, with a tool which cuts a plug 4 inches in diameter. The plug shall be withdrawn and ten of the discs which make up the plug shall be removed as follows:

(a) One disc from the second sheet from wrapper.
(b) Two discs, 1 inch deep.
(c) Three discs, 2 inches deep.
(d) Four discs, 3 inches deep.

Each roll shall be bored but once. The edge of the hole bored in the first roll shall be 2 inches from one end. The holes in the succeeding rolls shall be spaced at intervals of one-fifth the distance across the roll and staggered so as to take in the entire circumference of the roll. These discs shall be used for the tests specified in 4.4.6 through 4.4.12.

4.3.3.1.1.2 *For determination of physical properties.* A sheet 72 inches in length shall be cut from each roll before boring operation and subjected to the test specified in 4.4.2 to 4.4.5.

4.3.4 *Rejection.* If any sample fails to comply with any requirement, the lot shall be rejected.

4.4 Test methods and procedures.

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4.4.1 *Determination of moisture code No. 05.* The percentage of moisture shall be determined in accordance with Specification UU–P–31, method 240 or TAPPI method T412m. The percentage of moisture present in the original specimen shall be calculated as follows:

$$\text{Percentage of moisture present} = \frac{(A-B)\ 100}{A}$$

Where:

A=weight in grams (gms) of the original specimen.

B=weight in gms of the dried specimen.

The moisture factor (F) used as a correction factor to reduce "as is" test specimens to a dry basis for use in other determinations shall be calculated as follows:

$$\text{Moisture Factor (F)} = \frac{C}{W}$$

Where:

C=weight in gms of dried specimen.

W=weight in gms of original specimen.

The dried specimen shall be retained for the determination of either extractive matter in 4.4.8.

4.4.2 *Determination of density code No 06.* The density shall be determined in accordance with Specification UU–P–31, method 110 or TAPPI Method T411m, except that the calculation shall be made in ounces per square foot per 0.01 inch of thickness.

4.4.3 *Determination of thickness code No. 07.* The thickness shall be determined in accordance with Specification UU–P–31, method 173 or TAPPI Method T411m.

4.4.4 *Determination of weight code No. 08.* The weight shall be determined in accordance with Specification UU–P–31, method 110 or TAPPI Method T410m, except that calculation shall be made in ounces per square foot.

4.4.5 *Determination of bursting strength code No. 09.* The bursting strength shall be determined in accordance with Specification UU–P–31, method 112 or TAPPI Method T403m.

4.4.6 *Determination of viscosity code No. 10.* The viscosity shall be determined in accordance with the Cupriethylenediamine method in TAPPI code No. T230 SM–50.

4.4.7 *Determination of alpha cellluose code No. 11.* The percentage of alpha cellulose shall be determined in accordance with Specification UU–P–31, method 201 or TAPPI method T–235–M58.

4.4.8 *Determination of ether extractive matter code No. 12.* The specimen retained from the moisture determination 4.4.1 shall be transferred to a Soxhlet or similar extractor. Sufficient ether, to carry out the extraction shall be added, and the specimen shall be extracted on a steam bath for a maximum of 3 hours. The ether extract shall be poured into a tared 250-ml beaker. The flask and the extractor shall be individually rinsed with several small portions of ether, and the rinsing shall be added to the contents of the beaker. The beaker and its contents shall be placed on a steam bath in the hood, with filter paper under the beaker. The contents of the beaker shall be carefully evaporated. A gentle current of air, if available, may be used to aid in the evaporation. After the ether has evaporated, the beaker and residue shall be transferred to a drying oven maintained at 105°±2°C. for 1 hour. At the conclusion of this period, the beaker shall be cooled in a desiccator and weighed. The percentage of ether extractive matter shall be calculated on a moisture-free basis as follows:

$$\text{Percentage of ether extractive} = \frac{(A-B)\ 100}{W}$$

Where:

A=weight of beaker and residue in gm after evaporation.

B=weight of empty beaker in gm

W=weight in gm of the specimen (dried).

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4.4.9 *Determination of alkali soluble matter code No. 13.* A portion of the undried sample, of approximately 2.0 gm weighed to the nearest 0.1 milligram (mg) shall be transferred to a 250 ml Erlenmeyer flask. A volume of 100 ml of 1.92±0.10 Normal (N) (7.14±0.10 percent) sodium hydroxide solution shall be added to the flask. The flask shall be fitted with a rubber stopper through which passes a long glass tube (min. 36 in.) which serves as an air cooled reflux condenser. The flask shall be permitted to reflux for 3 hours at a temperature of 100°± 2°C. At the end of this period, the contents of the flask shall be poured into a 2-liter flask containing 1 liter of distilled water. The excess alkali present shall be neutralized using an excess of acetic acid solution. Blue litmus paper may be employed as an indicator. A clean, dry, prepared Gooch crucible shall be placed in a clean, dry, glass-stoppered weighing bottle. The crucible and the weighing bottle shall be weighed to the nearest 0.1 mg. The contents of the beaker shall be transferred to the Gooch crucible within 5 minutes. The cellulose residue in the Gooch shall be thoroughly washed with successive portions of hot water, alcohol, and ether. The crucible and contents shall be placed in an oven maintained at 105±2°C. for 2 hours. The crucible shall be placed in the weighing bottle, glass stopper and contents shall be cooled to room temperature in a desiccator. The stopper shall be replaced in the weighing bottle, and then the weighing bottle and its contents shall be weighed. The drying, cooling, and weighing procedure shall be repeated at hourly intervals until 2 successive weighings do not differ by more than 0.5 mg or there is a gain in weight. The percentage of alkali soluble matter shall be calculated on a moisture-free basis as follows:

$$\text{Percentage of alkali soluble matter} = \frac{(FA-B)\ 100}{AF}$$

Where:

A=weight in gm of original specimen.
B=weight in gm of dried residue.
F=moisture factor (see 4.4.1).

4.4.10 *Determination of ash code No. 14.* The percentage of ash present shall be determined in accordance with Specification UU-P-31, method 202 or TAPPI Method T413m. The percentage of ash shall be calculated on a moisture-free basis as follows:

$$\text{Percent of ash} = \frac{100\ A}{BF}$$

Where:

A=weight of ash in gm
B=weight of "as is" specimen.
F=moisture factor (see 4.4.1).

4.4.11 *Determination of lignin code No. 15.* The percentage of lignin shall be determined in accordance with Specification TAPPI code number T222-M-50. The percent of lignin present shall be calculated on a moisture-free basis as follows:

$$\text{Percent of lignin} = \frac{100\ A}{W\ F}$$

Where:

A=weight in gm of lignin found in specimen.
B=weight in gm of undried specimen.
F=moisture factor.

4.4.12 *Determination of silica code No. 16.* A portion of the sample, of approximately 40.0 gm shall be weighed to the nearest 1.0 mg and transferred to a porcelain crucible of adequate capacity. The crucible and contents shall be placed on a triangle supported on a tripod in a hood. The crucible shall be carefully and gently heated over a Bunsen burner until the woodpulp ignites and heat until a black, residue remains. The crucible shall then be transferred to a muffle maintained at a temperature of 525°±25°C. for one-half hour. At the end of this period, the crucible shall be removed and allowed to cool to room temperature in a desiccator. When cool, 5 ml of 72 percent perchloric acid shall be added and the crucible shall be heated to white fumes on an asbestos covered hotplate in a hood. The fuming shall be continued for a minimum of 10 minutes or until the crucible is dry. The crucible and contents shall then be cooled to room temperature 1 ml of concentrated hydrochloric acid shall be added followed by 50 ml of distilled water and the mixture shall be heated to boiling. The mix-

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ture shall be filtered through a Whatman No. 4 filter paper or equal. The crucible shall be rinsed with three 25-ml portions of hot distilled water each containing 5 drops of concentrated hydrochloric acid, letting one portion pass completely through the filter before adding the next. Any residue present in the crucible after ignition shall be completely transferred to the filter. The residue on the filter shall be washed with small portions of hot distilled water until the filtrate is neutral to litmus. The filter paper and residue shall be transferred to a tared platinum crucible which shall be gently heated until the filter paper is dry and then heated until the filter paper ignites. The filter paper be allowed to burn until completely charred. The crucible shall then be transferred to a muffle furnace maintained at 850°±50°C. for 1 hour. At the end of this period the crucible shall be transferred to a desiccator and cooled and then weighed. The crucible shall then be transferred to an asbestos covered hotplate and 5 ml of hydrofluoric acid followed by 10 drops of concentrated sulfuric acid shall be added to the residue in the crucible. The crucible and its contents shall be heated on the hotplate until dense white fumes are no longer emitted from its contents. The crucible shall be placed in a muffle furnace maintained at 850°±50°C. and its contents shall be reignited. At the end of ½ hour the crucible shall be removed from the furnace and placed in a desiccator. When cool, the crucible shall be removed from the desiccator and weighed. The percentage of silica shall then be calculated as follows:

$$\text{Percentage of silica} = \frac{(A-B)\ 100}{W\ F}$$

Where:

A = weight in gm of platinum crucible and contents after ignition and prior to addition of hydrofluoric acid.

B = weight in gm of platinum crucible and contents after addition of hydrofluoric acid and ignition.

W = weight in gm of specimen.

F = moisture factor (see 4.4.1).

4.4.13 *Determination of pentosans code No. 17.* Percentage of pentosans shall be determined in accordance with TAPPI method T223–M–48.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 *Level A.* Unless otherwise specified, cellulose woodpulp shall be furnished in rolls of approximately 30 inches wide maximum and 42 inches diameter maximum. Each roll shall be supported by a central core of approximately 2.75 inches inside diameter. Rolls shall be suitably restrained from unwinding. Each roll shall be completely wrapped and headed with at least one thickness of Kraft wrapping paper, grade B, 60-pound basis weight in accordance with Specification UU–P–268. The paper shall be sealed with Grade B adhesive in accordance with Specification MIL–A–140 or with 4-inch wide water-resistant tape complying with Specification UU–T–116 or 3-inch wide water-resistant tape complying with Specification PPP–T–76, centered along the end edge of the paper and applied across the full width of the roll.

5.1.2 *Level C.* Unless otherwise specified, rolls, supported on cores as specified in 5.1.1, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipments shall comply with Consolidated Freight Classification Rules and Container Specifications for rail shipment or National Motor Freight Rules and Container Specifications for truck shipments, as applicable.

5.2 Marking. Unless otherwise specified, markings shall be in accordance with MIL–STD–129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Initial production (see 4.2).

(c) Level of packing required (see 5.1).

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6.2 **Intended use.** The intended use of the cellulose woodpulp (sulfate) is in the manufacture of nitrocellulose.

6.3 **Inspection code number.** The two- and five-digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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